

Logically Equivalent
Equivalence Proofs
Predicates
Quantifiers
Translating English to Logic

Lecture-2

Logically Equivalent Predicates and Quantifiers



Logically Equivalent

- Two compound propositions p and q are logically equivalent if $p \leftrightarrow q$ is a tautology.
- We write this as $p \Leftrightarrow q$ or as $p \equiv q$ where p and q are compound propositions.
- Two compound propositions p and q are equivalent if and only if the columns in a truth table giving their truth values agree.
- This truth table show $\neg p \vee q$ is equivalent to $p \rightarrow q$.

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p	q	$\neg p$	$\neg p \vee q$	$p \rightarrow q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

De Morgan's Laws

$$(a) \neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$(b) \neg(p \vee q) \equiv \neg p \wedge \neg q$$

p	q	$\neg p$	$\neg q$	$(p \vee q)$	$\neg(p \vee q)$	$\neg p \wedge \neg q$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

De Morgan's Laws

$$(a) \neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$(b) \neg(p \vee q) \equiv \neg p \wedge \neg q$$

Logical Equivalence Laws

Identity Laws	$p \wedge T \equiv p$	$p \vee F \equiv p$
Domination Laws	$p \vee T \equiv T$	$p \wedge F \equiv F$
Idempotent Laws	$p \vee p \equiv p$	$p \wedge p \equiv p$
Double Negation law	$\neg(\neg p) \equiv p$	
Negation Laws	$p \vee \neg p \equiv T$	$p \wedge \neg p \equiv F$
Commutative laws	$p \vee q \equiv q \vee p$	$p \wedge q \equiv q \wedge p$

Logical Equivalence Laws

Associative laws	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	$(p \vee q) \vee r \equiv p \vee (q \vee r)$
Distributive laws	$(p \vee (q \wedge r)) \equiv (p \vee q) \wedge (p \vee r)$	$(p \wedge (q \vee r)) \equiv (p \wedge q) \vee (p \wedge r)$
Absorptions laws	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$

More Logical Equivalences

Logical Equivalences Involving Conditional Statements.

$$p \rightarrow q \equiv \neg p \vee q$$

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

$$p \vee q \equiv \neg p \rightarrow q$$

$$p \wedge q \equiv \neg(p \rightarrow \neg q)$$

$$\neg(p \rightarrow q) \equiv p \wedge \neg q$$

$$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$$

$$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$$

$$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$$

$$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$$

Logical Equivalences Involving Biconditional Statements.

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

$$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$$

$$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$$

$$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$$

Equivalence Proofs

Question :show that $\neg(p \vee (\neg p \wedge q))$ is logically equivalent to $\neg p \wedge \neg q$.

$$\begin{aligned}\neg(p \vee (\neg p \wedge q)) &\equiv \neg p \wedge \neg(\neg p \wedge q) \text{ by second De Morgan Law} \\ &\equiv \neg p \wedge (p \vee \neg q) \text{ by the double negation Law} \\ &\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) \text{ by Distributive Law} \\ &\equiv F \vee (\neg p \wedge \neg q) \\ &\equiv (\neg p \wedge \neg q) \text{ BY Identity law}\end{aligned}$$

Equivalence Proofs

Question : Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

$$\begin{aligned}(p \wedge q) \rightarrow (p \vee q) &\equiv \neg(p \wedge q) \vee (p \vee q) \\ &\equiv \neg p \vee \neg q \vee p \vee q \quad \text{negation Law} \\ &\equiv (\neg p \vee p) \vee (q \vee \neg q) \\ &\equiv T \vee T \\ &\equiv T\end{aligned}$$

Equivalence Proofs

Question : Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

$$\begin{aligned}(p \wedge q) \rightarrow (p \vee q) &\equiv \neg(p \wedge q) \vee (p \vee q) \\ &\equiv \neg p \vee \neg q \vee p \vee q \quad \text{negation Law} \\ &\equiv (\neg p \vee p) \vee (q \vee \neg q) \\ &\equiv T \vee T \\ &\equiv T\end{aligned}$$

Precedence of Logical Operators

Operator	Precedence
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

$p \vee q \rightarrow \neg r$ is equivalent to $(p \vee q) \rightarrow \neg r$

If the intended meaning is $p \vee (q \rightarrow \neg r)$
then parentheses must be used.

Find the truth value of $p \vee q \rightarrow \neg r$

- Predicates

• Introducing Predicate Logic

- Predicate logic uses the following new features:
 - Variables: x, y, z
 - Predicates: $P(x), M(x)$
 - Quantifiers (*to be covered in a few slides*):

- *Propositional functions* are a generalization of propositions.
 - They contain variables and a predicate, e.g., $P(x)$
 - Variables can be replaced by elements from their *domain*.
- Propositional functions become propositions (and have truth values) when their variables are each replaced by a value from the *domain* (or *bound* by a quantifier, as we will see later).

- The statement $P(x)$ is said to be the value of the propositional function P at x . For example, let $P(x)$ denote “ $x > 0$ ” and the domain be the integers.

Then:

$P(-3)$ is false.

$P(0)$ is false.

$P(3)$ is true.

- Often the domain is denoted by U . So in this example U is the integers.

- Examples of Propositional Functions

Let “ $x + y = z$ ” be denoted by $R(x, y, z)$ and U (for all three variables) be the integers. Find these truth values:

$R(2, -1, 5)$

Solution: F

$R(3, 4, 7)$

Solution: T

$R(x, 3, z)$

Solution: Not a Proposition

Now let “ $x - y = z$ ” be denoted by $Q(x, y, z)$, with U as the integers. Find these truth values:

$Q(2, -1, 3)$

Solution: T

$Q(3, 4, 7)$

Solution: F

$Q(x, 3, z)$

Solution: Not a Proposition

- Quantifiers

- Quantifiers:
- We need *quantifiers* to express the meaning of English words including *all* and *some*:
 - “All men are Mortal.”
 - “Some students have laptop”

- The two most important quantifiers are:
 - *Universal Quantifier*, “For all,” symbol: $\forall$
 - *Existential Quantifier*, “There exists,” symbol: $\exists$

- We write as in $\forall x P(x)$ and $\exists x P(x)$.
- $\forall x P(x)$ asserts $P(x)$ is true for every x in the *domain*.
- $\exists x P(x)$ asserts $P(x)$ is true for some x in the *domain*.
- The quantifiers are said to bind the variable x in these expressions.

- Quantifiers

Universal Quantifier:

- $\forall x P(x)$ is read as “For all x , $P(x)$ ” or “For every x , $P(x)$ ”

Examples:

- 1) If $P(x)$ denotes “ $x > 0$ ” and U is the integers, then $\forall x P(x)$ is false.
- 2) If $P(x)$ denotes “ $x > 0$ ” and U is the positive integers, then $\forall x P(x)$ is true.
- 3) If $P(x)$ denotes “ x is even” and U is the integers, then $\forall x P(x)$ is false.

Existential Quantifier:

- $\exists x P(x)$ is read as “For some x , $P(x)$ ”, or as “There is an x such that $P(x)$,” or “For at least one x , $P(x)$.”

Examples:

1. If $P(x)$ denotes “ $x > 0$ ” and U is the integers, then $\exists x P(x)$ is true. It is also true if U is the positive integers.
2. If $P(x)$ denotes “ $x < 0$ ” and U is the positive integers, then $\exists x P(x)$ is false.
3. If $P(x)$ denotes “ x is even” and U is the integers, then $\exists x P(x)$ is true.

Properties of Quantifiers:

- The truth value of $\exists x P(x)$ and $\forall x P(x)$ depend on both the propositional function $P(x)$ and on the domain U .
- **Examples:**
 1. If U is the positive integers and $P(x)$ is the statement " $x < 2$ ", then $\exists x P(x)$ is true, but $\forall x P(x)$ is false.
 2. If U is the negative integers and $P(x)$ is the statement " $x < 2$ ", then both $\exists x P(x)$ and $\forall x P(x)$ are true.
 3. If U consists of 3, 4, and 5, and $P(x)$ is the statement " $x > 2$ ", then both $\exists x P(x)$ and $\forall x P(x)$ are true. But if $P(x)$ is the statement " $x < 2$ ", then both $\exists x P(x)$ and $\forall x P(x)$ are false.

- Quantifiers

Precedence of Quantifiers

- The quantifiers $\forall$ and $\exists$ have higher precedence than all the logical operators.
- For example, $\forall x P(x) \vee Q(x)$ means $(\forall x P(x)) \vee Q(x)$
- $\forall x (P(x) \vee Q(x))$ means something different.

Translating from English to Logic

Example 1: Translate the following sentence into predicate logic:
“Every student in this class has taken a course in Java.”

Solution 1:

First decide on the domain U .

Solution 1: If U is all students in this class, define a propositional function $J(x)$ denoting “ x has taken a course in Java” and translate as $\forall x J(x)$.

Translating from English to Logic

Example 2: Translate the following sentence into predicate logic:
“Some student in this class has taken a course in Java.”

Solution 1:

If U is all students in this class, translate as

$$\exists x J(x)$$

Equivalences in Predicate Logic

- Statements involving predicates and quantifiers are *logically equivalent* if and only if they have the same truth value

The notation $S \equiv T$ indicates that S and T are logically equivalent.

Example: $\forall x \neg \neg S(x) \equiv \forall x S(x)$

De Morgan's Laws for Quantifiers

- The rules for negating quantifiers are
- $\neg \exists x p(x) \equiv \forall x \neg p(x)$
- $\neg \forall x p(x) \equiv \exists x \neg p(x)$

The notation $S \equiv T$ indicates that S and T are logically equivalent.
Example: $\forall x \neg \neg S(x) \equiv \forall x S(x)$

Nested Quantifiers

Example: “Every real number has an inverse” is

$$\forall x \exists y (x + y = 0)$$

where the domains of x and y are the real numbers.

Order of Quantifiers: Examples:

1. Let $P(x,y)$ be the statement “ $x + y = y + x$.” Assume that U is the real numbers. Then $\forall x \forall y P(x,y)$ and $\forall y \forall x P(x,y)$ have the same truth value.
2. Let $Q(x,y)$ be the statement “ $x + y = 0$.” Assume that U is the real numbers. Then $\forall x \exists y P(x,y)$ is true, but $\exists y \forall x P(x,y)$ is false

Questions on Order of Quantifiers :

Example 1: Let U be the real numbers,

Define $P(x,y) : x \cdot y = 0$

What is the truth value of the following:

1. $\forall x \forall y P(x,y)$

Answer: False

2. $\forall x \exists y P(x,y)$

Answer: True

3. $\exists x \forall y P(x,y)$

Answer: True

4. $\exists x \exists y P(x,y)$

Answer: True